

# NAVEEN RAJARAPU

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## SUMMARY

Applied AI Engineer with 3+ years of experience building enterprise AI assistants, retrieval-based knowledge systems, structured extraction pipelines, computer vision monitoring solutions, and healthcare machine learning workflows. Strong background in Python, FastAPI, SQL, Docker, Azure, computer vision, and modern LLM application development, with hands-on experience across conversational AI, speech-to-text pipelines, model evaluation, synthetic data generation, and privacy-aware training.

## TECHNICAL SKILLS

**Generative AI & NLP:** LLM Applications, Retrieval-Augmented Generation (RAG), Conversational AI, Agentic Workflows, Prompt Engineering, LangChain, LangGraph, LangSmith, ChromaDB, BM25, Embeddings, OpenAI API, Groq API, Whisper, Transformers, BERT, LoRA, QLoRA, Hugging Face PEFT

**Backend, Deployment & Data:** Python, FastAPI, Flask, REST APIs, SQL, PostgreSQL, SQLAlchemy, JWT Authentication, Docker, GitHub Actions CI/CD, Azure, AWS S3, Render

**Machine Learning & Computer Vision:** PyTorch, TensorFlow, Keras, scikit-learn, CNNs, LSTM, GANs, Federated Learning, OpenCV, Object Detection, Video Analytics, Tracking, Time-Series Modeling

**Evaluation & Reliability:** MLflow, RAGAS, DeepEval, Guardrails AI, Hallucination Detection, BLEU, ROUGE, BERTScore, LLM-as-Judge

**Tools:** Git, Linux, MATLAB

## PROFESSIONAL EXPERIENCE

**AI Engineer - Medical AI & Deep Learning** | Lawrence Technological University, MI March 2025 – Present

- Worked on the development of an AI-driven monitoring workflow aimed at identifying signs of pacemaker malfunction before device failure, using ECG signal analysis to study pacing patterns, failure indicators, and clinically meaningful changes across 5+ malfunction scenarios.
- Researched pacemaker behavior, device failure modes, and ECG waveform characteristics to define a practical modeling approach for early detection, with the broader goal of supporting more proactive and reliable monitoring for patients with implanted cardiac devices.
- Built data preparation and experimentation workflows for ECG-based classification, including signal processing, dataset organization, and benchmarking of 4+ machine learning and deep learning approaches across different malfunction categories.
- Evaluated CNN-based models for waveform and morphology recognition, using them as the primary approach for detecting pacemaker-related abnormalities from ECG records and comparing performance across alternative model configurations.
- Explored LSTM-based and sequence-oriented models for time-series analysis to study how temporal ECG behavior could improve the ability to identify developing issues and support earlier prediction of likely pacemaker failure patterns.
- Addressed limited and imbalanced clinical data by implementing a GAN-based synthetic ECG generation pipeline, increasing minority class representation by approximately 3× and improving downstream model development for rare pacemaker malfunction patterns.
- Implemented federated learning experiments to explore privacy-aware model training on sensitive healthcare data, supporting a workflow better aligned with real-world medical constraints around data sharing, cross-site collaboration, and regulated environments.
- Tracked experiments across 20+ model runs, compared model results, and contributed to research documentation, helping translate technical findings into a more applied healthcare AI workflow focused on early risk detection and model reliability.

**AI Engineer** | R4IT Labs, India May 2022 – June 2024

- Delivered applied AI solutions across two major client programs in insurance operations and construction monitoring, building systems that combined retrieval-based assistants, speech and text pipelines, structured data workflows, and computer vision-based site intelligence.
- **Insurance knowledge assistant:** Built an internal AI assistant for a large insurance organization supporting multiple associations, enabling employees to ask questions over 10,000+ policy documents, operational updates, process documents, and internal knowledge sources through a retrieval-based workflow with grounded references.
- Designed the document ingestion and retrieval pipeline for the internal assistant, including content preparation, chunking, embeddings, retrieval, and response generation, so users could access relevant information quickly without manually searching across multiple internal resources.
- **Call recording intelligence workflow:** Developed a voice-to-structured-data pipeline to process 200+ customer interaction recordings, using Azure and Whisper-based transcription followed by LLM-driven information extraction to convert conversations into summaries and database-ready structured fields.

- Built a review-friendly workflow for extracted call data so users could verify outputs, complete missing fields manually when required, and store validated information in the database for downstream retrieval, reporting, and operational use, reducing manual data entry effort by an estimated 35–40%.
- **Database assistant:** Implemented a conversational assistant over structured data that retrieved exact records, surfaced relevant columns based on user intent, and asked follow-up clarification questions when queries were incomplete or multiple matching values were present.
- **Internal website copilot:** Built a private AI copilot for the client’s internal web platform to help employees keep up with frequent content changes, process updates, and newly published information without relying on public copilots or external AI tools.
- Designed the internal copilot workflow to interpret newly updated content and surface concise, relevant explanations to employees, improving awareness of day-to-day operational changes and reducing the effort required to continuously monitor internal portals.
- **Construction site monitoring system:** Implemented a computer vision solution for a construction client to monitor live site activity, track 5+ heavy equipment types, count vehicle entries and exits, and estimate active versus idle durations from surveillance footage.
- Built the site-monitoring workflow to improve visibility into whether planned machinery was actually being used in the field, giving owners and operations teams a clearer view of underutilization, worksite activity, and execution gaps across the project.
- Diagnosed and resolved a critical accuracy issue in activity-duration measurement by redesigning the frame-comparison logic used for temporal analysis, reducing idle-time estimation error by approximately 30% and strengthening the reliability of the production workflow.
- Collaborated with stakeholders and delivery teams across implementation phases, translating operational needs into practical AI workflows that supported internal users, customer-facing staff, and management teams in day-to-day decision-making.

## PROJECTS

**Medical Evidence Agent-Production-Oriented AI System** *Python, LangGraph, FastAPI, Guardrails AI, Pydantic, PostgreSQL, Docker, AWS S3, GitHub Actions, LangSmith, MLflow*

- Built a medical evidence assistant designed to answer document-grounded questions using a multi-stage workflow for retrieval, reasoning, and safety validation, improving response reliability for knowledge-intensive medical queries across a corpus of 5,000+ indexed documents.
- Developed the backend using FastAPI, PostgreSQL, AWS S3, Docker, and LangSmith, with structured output validation, query tracking, and experiment logging to support maintainability, traceability, and repeatable evaluation.
- Evaluated system quality using RAGAS, grounding checks, and safety-focused validation logic, improving faithfulness scores by approximately 20% and reducing unsupported answers in production-oriented testing scenarios.

**PersonaWeave-Personalized Dialogue Generation Study** *Python, LoRA, Hugging Face PEFT, LangChain, RAG, BLEU, ROUGE, BERTScore*

- Built an NLP pipeline to compare prompt engineering, retrieval-based generation, and LoRA fine-tuning for personality-conditioned dialogue generation using real WhatsApp conversation data across 14+ participants and conversational settings.
- Designed an evaluation workflow using BLEU, ROUGE, and BERTScore to compare response quality across methods and analyze tradeoffs between retrieval-based personalization and parameter-efficient fine-tuning approaches.
- Documented experiments and outputs to support a reproducible comparison of low-resource personalization strategies for conversational AI systems.

## EDUCATION

**Lawrence Technological University** | MS in Artificial Intelligence  
GPA: 3.72 / 4.0

*Expected May 2026*